

# FitBuddy - AI Fitness Plan Generator using Gemini Models

## Brainstorming & Ideation

|              |                                                                 |                  |                                |
|--------------|-----------------------------------------------------------------|------------------|--------------------------------|
| Date         | 29 September 2026                                               | Team ID          | SWTID-2026-5364                |
| Project Name | FitBuddy - AI Fitness Plan Generator using Gemini Models        | Maximum Marks    | 3 Marks                        |
| Team Members | D.Kalaiaarasi, J.Devipriya, E.Dhanalakshmi, S.Gopika, N.Sharuli | Technology Focus | FastAPI, Google Gemini, SQLite |

## Define Problem Statements

The team translated the empathy findings and brainstorming results into specific problem statements that the FitBuddy solution addresses.

| Problem Area       | Observed User Need                                                               | FitBuddy Response                                                    |
|--------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Personalization    | Generic plans may not match an individual's goal or preferred workout intensity. | Gemini generates a personalized 7-day workout plan from user inputs. |
| Nutrition Guidance | Users may need simple nutrition guidance connected to their fitness goal.        | Gemini Flash provides goal-oriented nutrition tips.                  |
| Adaptability       | A plan may need changes after the user tries it and provides feedback.           | Feedback is used to generate an updated workout plan.                |
| Continuity         | Users need their profile and generated plans to persist for later review.        | SQLite/SQLAlchemy stores user and plan information.                  |

**Core Problem Statement:** Fitness users, especially beginners and busy users, need an accessible way to receive personalized workout and nutrition guidance that can adapt to their feedback, because generic fitness routines can be difficult to personalize, follow, and maintain consistently.